

COURSE SYLLABUS

AI Essentials

COURSE NUMBER

AIE-101

DURATION

1 day · 4 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This one-day course teaches the foundational skills required to leverage modern artificial intelligence tools to boost professional productivity. Students will learn the mechanics of prompt engineering, automated professional writing, basic data analysis, and workflow optimization. Through a series of hand-on browser-based exercises, participants will gain practical, vendor-neutral skills to solve real-world workplace challenges using AI.

DELIVERY FORMAT OPTIONS

Standard Instructor-Led Training (ILT). The instructor teaches each module, leads explanations and discussions, and guides students through the standard module labs using the published module timing.

Bootcamp / Lab-Centered Delivery. When Bootcamp Mode capstone labs are included, they provide an alternative delivery configuration, not additional required ILT lecture time. In this configuration, students primarily work through module labs and capstone lab scenarios while the trainer acts as a facilitator, answering questions, clarifying concepts, and helping students work through lab challenges.

Training providers may choose either the normal ILT model or the Bootcamp/lab-centered model based on audience readiness, delivery goals, and available class time.

AT COURSE COMPLETION

After completing this course, you will be able to:

Navigate modern web-based AI interfaces and execute structured prompt engineering principles.

Draft, summarize, edit, and optimize professional communications for targeted corporate audiences.

Clean, interpret, and extract quantitative patterns and qualitative insights from raw data sources.

Formulate structured action plans and design workflows to automate repetitive administrative procedures.

PREREQUISITES

- Basic computer literacy and comfort navigating web applications
 - Familiarity with standard productivity tools like word processors and spreadsheets
 - An active web browser and internet connectivity
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COURSE OUTLINE

MODULE 1 Foundations of Generative AI and Prompt Engineering

105 min

Establish a foundational understanding of generative AI and master the core mechanics of effective prompt design. Learn how to construct, test, and refine prompts to get high-quality, predictable outputs.

LESSONS

Lesson 1: Demystifying Large Language Models and Generative AI

Lesson 2: The Anatomy of an Effective AI Prompt

Lesson 3: Structuring Prompts with Role, Context, and Constraints

Lesson 4: Leveraging Zero-Shot and Few-Shot Learning Frameworks

Lesson 5: Recognizing and Mitigating AI Hallucinations

LAB EXERCISES

Crafting Precision Prompts with Role, Context, and Few-Shot Examples

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Explain the core mechanics and limitations of Large Language Models (LLMs)

Construct highly effective AI prompts using roles, context, instructions, and constraints

Apply few-shot prompting techniques to establish specific output standards

Identify common AI hallucinations and apply mitigation strategies

MODULE 2 Text Generation and Professional Writing Workflows

105 min

Discover how to accelerate writing and communication workflows using AI tools. Learn to draft, edit, and summarize complex business correspondence while maintaining human-in-the-loop control.

LESSONS

Lesson 1: Drafting Professional Correspondence and Reports

Lesson 2: Summarizing Lengthy Documents and Extracting Key Action Items

Lesson 3: Adapting Tone, Style, and Format for Diverse Audiences

Lesson 4: Iterative Editing and Collaborating with AI Assistants

Lesson 5: Verifying AI Facts and Cross-Referencing Sources

LAB EXERCISES

Refining Professional Communications with AI Assistants

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Generate professional business communications across multiple styles and formats

Summarize lengthy documents, reports, and meeting transcripts into actionable summaries

Adapt the tone, style, and structure of pre-existing documents for target audiences

Implement a structured human-in-the-loop review process to verify facts and style constraints

MODULE 3 Data Analysis and Information Synthesis

105 min

Gain the ability to interpret, analyze, and visualize data trends using general-purpose AI helpers. Learn to extract meaning from qualitative and quantitative data inputs.

LESSONS

Lesson 1: Formatting and Cleaning
Tabular Data with AI

Lesson 2: Extracting Trends and
Insights from Unstructured
Text

Lesson 3: Generating Formulas and
Code Snippets for Data
Manipulation

Lesson 4: Visualizing Data Trends using
AI-Guided Recommendations

Lesson 5: Critical Assessment of AI-
Generated Data
Interpretations

LAB EXERCISES

Analyzing and Synthesizing Customer
Feedback with AI

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Clean and organize unstructured qualitative data using conversational AI instruction sets

Identify macro-level trends and insights from structured customer or system data tables

Generate spreadsheet formulas and programmatic snippets to manipulate local documents

Evaluate AI-generated data summaries critically to detect statistical and logical errors

MODULE 4 Creative Problem Solving and Task Automation

105 min

Harness AI as a brainstorming partner and structural architect. Learn how to outline complex workflows, break down tasks, and configure logical templates to automate standard daily procedures.

LESSONS

- Lesson 1:** Brainstorming and Ideation Techniques with Generative Models
- Lesson 2:** Designing Multi-Step Workflows for Repetitive Tasks
- Lesson 3:** Utilizing AI for Basic Project Planning and Task Breakdown
- Lesson 4:** Selecting the Right AI Tool for Specific Professional Challenges
- Lesson 5:** Establishing a Continuous Learning Feedback Loop with AI

LAB EXERCISES

- Automating Project Workflows with AI Planning Models

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Leverage AI systems for lateral thinking, brainstorming, and divergent idea generation
- Deconstruct a broad project objective into a logical step-by-step implementation sequence
- Construct repeatable templates and workflow frameworks using conversational automation planners
- Select the appropriate AI platform, configuration, and interface pattern for diverse task sets

BOOTCAMP MODE — CAPSTONE LABS

Capstone labs are included for Bootcamp/lab-centered delivery. They are not additional required lecture time for Standard Instructor-Led Training. Use them when the client chooses a facilitated lab delivery model where the trainer coaches students through integrated practice scenarios.

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

Constructing multi-tier structured prompts using role, context, and constraint parameters

Summarizing unstructured qualitative data and formatting tone for professional delivery

Cleaning raw feedback datasets and synthesizing actionable recommendations

Designing a multi-step task automation plan to streamline business workflows

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.